KinetiQ

Non-Functional Requirements

Milestone 2 · Lifecycle Architecture · PEI 2026/2027, Universidade de Aveiro

KinetiQ non-functional requirements, aligned with the final MS2 architecture (revised on 31 Oct 2026) and with the functional requirements (FR01 to FR26).

Usability and Accessibility

ID	Requirement	Description
NFR01	Learnability	A Fleet Operator with no prior training must be able to check the fleet status and KPIs (FR09) from the home screen.
NFR02	Navigation efficiency	The fleet status must be visible on the home screen without further navigation, and the details of any bike must be at most two clicks away from the map (FR17). **To be confirmed:** depends on FR17, which the team has flagged as uncertain.
NFR03	Readability while riding	During a trip, the mobile app must show speed, battery and remaining range (FR21) on a single screen, in large text, without requiring interaction while moving.
NFR04	System feedback	The interface must show when live data is stale or the connection is lost, and resume automatically when the connection is restored.
NFR05	Error prevention	Actions with operational impact, such as retiring a bike (FR05, FR06), must require confirmation.
NFR06	Accessibility	The main flows (authentication, fleet status, bike details and alerts) must meet WCAG 2.2 level AA, including contrast and keyboard navigation, and must not convey status by colour alone. Map views must have a list alternative. Compliance is checked with an automated tool (e.g. Lighthouse or axe) and a manual keyboard test.
NFR07	Visual consistency	The web and mobile apps must share the same colours, typography and terminology.
NFR08	Role-based interface	The interface must show each role (Operator, Technician, Analyst and Rider, FR24) only the features it is allowed to use.

Interface Constraints

ID	Requirement	Description
NFR09	Interface technologies	The web app must be built with React and the mobile app with Flutter.
NFR10	Data access	The interfaces must access data only through the Reverse Proxy and the API Gateway, over HTTPS (REST) for requests and WSS for live data, with no direct access to the MQTT Broker, Kafka or the databases.
NFR11	Screen adaptation	The web back-office must be usable on desktop and tablet screens.
NFR12	Geographic filtering	Map views must allow filtering by the geographic zones defined in the system (FR26).

Performance and Capacity

ID	Requirement	Description
NFR13	API response time	95% of synchronous read requests to the REST API (e.g. bike status or KPI queries) must complete in under 200 ms under normal load.
NFR14	Event pipeline latency	The time from the arrival of a telemetry packet at the MQTT Broker to its persistence in InfluxDB and broadcast by the WebSocket Service must not exceed 500 ms. **To be confirmed:** whether this value is a target or an acceptance criterion.
NFR15	Throughput	The ingestion pipeline must continuously process at least 500 messages per second, without losing events and while keeping the NFR14 latency. **To be confirmed:** whether this value is a target or an acceptance criterion.

Scalability and Availability

ID	Requirement	Description
NFR16	Horizontal scalability	Services that consume and process events must be decoupled and stateless, so they can scale horizontally by adding instances to the same Kafka consumer group.
NFR17	Availability	The core telemetry ingestion services and the back-office must achieve at least 99% availability under normal operation.
NFR18	Recoverability and fault tolerance	If a database or downstream service fails temporarily, Kafka must retain messages according to the configured retention policy, and consumers must resume from the last committed offset, with no data loss.

Security and Data Integrity

ID	Requirement	Description
NFR19	Encrypted communication	All external traffic between the clients (web and mobile) and the backend must be encrypted with HTTPS (TLS 1.3) and WSS, and IoT traffic with MQTTS (TLS).
NFR20	Secure credential storage	Credentials are stored and managed only by Keycloak, hashed and salted. No other service stores passwords or implements its own authentication logic.
NFR21	Telemetry data integrity	Devices (ESP32 and simulators) must authenticate with the MQTT Broker before publishing. Physically impossible readings (e.g. sudden GPS jumps, battery above 100% or negative speeds) must be isolated in the Dead-Letter Queue and never reach InfluxDB.

Maintainability and Interoperability

ID	Requirement	Description
NFR22	API interoperability	All public and integration APIs must follow the OpenAPI 3.0 specification and return JSON.
NFR23	Layer segregation	Databases and internal Kafka topics must not be reachable by external clients; all communication goes through the Reverse Proxy and the API Gateway.

ID	Requirement	Description
NFR24	Modularity and containerisation	All backend services (MQTT Broker, Kafka and connectors, processing services, APIs, ML and BI modules, Keycloak and databases) must run in Docker containers declared with Docker Compose, so that development and test environments are easy to replicate.

Open points

Points not yet decided by the team that affect these requirements.

Requirement	Open point
NFR14, NFR15	Are 500 ms and 500 messages/s targets or MVP acceptance criteria?
NFR02	Depends on FR17 (real-time map), which the team has flagged as uncertain.